

RESEARCH PROFILE

Organic and process chemist with Ph.D. training in natural products isolation and characterization, continuous-flow synthesis, gas-liquid photochemistry, automated reaction platforms, and scalable pharmaceutical process development. Experienced in translating laboratory chemistry into reproducible, data-rich, and automation-enabled workflows using reactor design, Python-controlled hardware, pump calibration, and rigorous HPLC/LC-MS/NMR analysis. Research interests align with enabling technologies for synthesis, including flow chemistry, photochemistry, electrosynthesis, catalysis, process intensification, and sustainable reaction development.

RESEARCH FOCUS AND FIT

- Continuous-flow photochemistry and gas-liquid reaction engineering for natural product modification; focusing on artemisinin synthesis from its biological precursor, dihydroartemisinic acid, emphasizing optimal oxygen delivery, residence-time control, photosensitizer loading, acid-catalyzed rearrangement, downstream processing to eliminate byproducts and ensure product purity.
- Automated synthesis platforms built from open-source hardware, Python interfaces, peristaltic/syringe pumps, motor drivers, mass-flow controllers, I2C devices, and sensors for reproducible small-molecule reaction execution.
- Emerging electroorganic synthesis interests focused on reactor characterizations, electrode geometry, diffusion-layer control, gas evolution, mass transport, and scale-up in batch and flow reactor formats.

EDUCATION

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| University of Connecticut - Storrs, CT | Expected Dec 2026 |
| <i>Ph.D. in Chemistry, Organic Chemistry GPA: 3.8/4.0</i> | |
| <ul style="list-style-type: none">• Doctoral research centered on continuous-flow artemisinin synthesis, flow-platform development, automated experimentation, and artemisinin-inspired analogue synthesis. | |
| University of Nairobi - Nairobi, Kenya | Sep 2018 - Sep 2022 |
| <i>M.Sc. in Chemistry, Organic Chemistry GPA: 3.9/4.0</i> | |
| <ul style="list-style-type: none">• Isolated and characterized bioactive naphthols and naphthoquinones from <i>Pentas</i> species and related medicinal plants; integrated chromatography, 1D/2D NMR, LC-MS, SAR analysis, and computational docking against trypanothione reductase. | |
| Kyambogo University - Kampala, Uganda | Aug 2013 - Dec 2016 |
| <i>B.Sc. in Education, Chemistry and Mathematics GPA: 4.34/5.0</i> | |
| <ul style="list-style-type: none">• Final project: analysis of heavy metals in soil and water samples around school laboratories. | |

RESEARCH AND PROFESSIONAL EXPERIENCE

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| Pfizer - Groton, CT | Jun 2026 - Aug 2026 |
| <i>Graduate Chemistry Intern</i> | |
| <ul style="list-style-type: none">• Designed and developed a flow scale-relevant methodology to enable efficient and selective decarboxylative acetoxylation of peptides.• Integrating electrochemistry with flow concepts to replace stoichiometric reagents with electrons for more sustainable synthesis at multigram scale.• Performing reaction monitoring and side-product analysis to tune selectivity, robustness, and process decision-making.• Developed enabling software toolkit for precision control of fluid handlers and electrochemical modules. | |

University of Connecticut - Storrs, CT**Sep 2022 - Present***Graduate Research Assistant, Gilmore Laboratory*

- Designed, optimized, and scaled a continuous-flow pharmaceutical process for converting dihydroartemisinic acid to artemisinin, integrating photooxidation, oxygen mass transfer, acid-catalyzed rearrangement, residence-time control, and crystallization; achieved high-purity artemisinin in optimized runs.
- Built a modular automated flow-chemistry platform using Raspberry Pi, Python, peristaltic pumps, motor drivers, printed circuit assemblies, sensors, and mass-flow control to streamline small-molecule synthesis and improve reproducibility.
- Developed Python-based control and calibration workflows for reagent delivery, pump correction, flow-rate verification, residence-time calculations, and safer process execution.
- Designed and synthesized artemisinin derivatives, including ether-linked, triazole-linked, and linker-free dimeric scaffolds, to explore stability, payload, and potential antimalarial/anticancer applications.
- Established analytical workflows using HPLC-UV, LC-MS, 1D/2D NMR, IR, and UV-Vis to quantify conversion, purity, residual starting materials, and product identity.
- Contributed to collaborative flow-synthesized quantum dot research, achieving substantially improved performance relative to batch through controlled reaction engineering.
- Applied GMP-aligned documentation, safety protocols, and data-traceability practices; mentored three undergraduate researchers in laboratory technique, synthesis workflows, and safety.

Busitema University, Natural Products Research & Innovation Center - Mbale, Uganda**Sep 2020 - Jan 2021***Research Assistant*

- Developed extraction and purification workflows for bioactive phytochemicals from medicinal plants guided by ethnopharmacology and indigenous knowledge.
- Characterized compounds by 1D/2D NMR and LC-MS and correlated structure with biological activity in a multidisciplinary natural-products discovery program.
- Coordinated project administration, documentation, and team workflows supporting development activities for the natural-product-derived therapeutic syrup TAZCOV.

TECHNICAL SKILLS

Enabling technologies: Continuous-flow chemistry, gas-liquid photochemistry, electrosynthesis concepts, process intensification, reactor design, residence-time control, scale-up thinking, reaction optimization.

Synthetic chemistry: Multistep organic synthesis, pharmaceutical process chemistry, natural product derivatization, artemisinin chemistry, CuAAC, etherification, amidation, dimer synthesis, purification, chromatography.

Automation and instrumentation: Python, Raspberry Pi, I2C devices, peristaltic/syringe pumps, motor drivers, MFCs, sensors, circuit boards, pump calibration, data acquisition, experimental automation.

Analytical and computational: HPLC-UV, LC-MS, NMR, IR, UV-Vis, Reaxys, SciFinder, Schrödinger Maestro, Avogadro, BIOVIA, data analysis.

LEADERSHIP AND SERVICE

**International Christian Fellowship, University of
Connecticut - Storrs, CT**

May 2024 - Jan 2026

President

- Led executive and membership meetings, coordinated events and prayer meetings, represented the organization at official functions, and managed relationships with officers, volunteers, and campus partners.

LANGUAGES

English (fluent); Luganda (native); Lusoga (native); Swahili (conversational).

REFEREES

Professor Kerry Gilmore

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Professor Christian Brueckner

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